

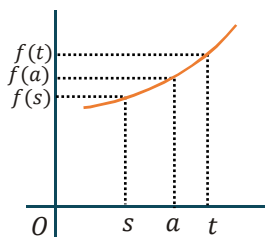
03

Monotonicity

MONOTONICITY (INTRODUCTION):

Increasing / Decreasing / Strictly Increasing / Strictly Decreasing Nature of a Function at a Point:

I. Increasing at $x = a$:



If $f(s) \leq f(a) \leq f(t)$ whenever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be increasing at $x = a$.

(1) When ' a ' be left end of the interval

$$f(a) \leq f(x) \forall x \in (a, a + h) \cap D_f \text{ for some } h > 0$$

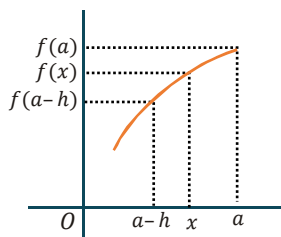
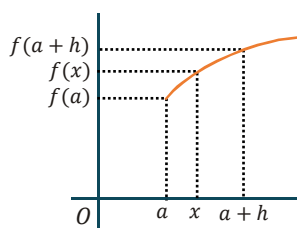
$\Rightarrow f$ is increasing at $x = a$.

(2) When ' a ' be right end of the interval

$$f(x) \leq f(a) \forall x \in (a - h, a) \cap D_f \text{ for some } h > 0$$

$\Rightarrow f$ is increasing at $x = a$.

II. Strictly Increasing at $x = a$:



If $f(s) < f(a) < f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be strictly increasing at $x = a$.

(1) When ' a ' be left end of the interval

$$f(a) < f(x) \forall x \in (a, a + h) \cap D_f \text{ for some } h > 0$$

$\Rightarrow f$ is strictly increasing at $x = a$.

(2) When ' a ' be right end of the interval

$$f(x) < f(a) \forall x \in (a - h, a) \cap D_f \text{ for some } h > 0$$

$\Rightarrow f$ is strictly increasing at $x = a$.

III. Decreasing at $x = a$:

If $f(s) \geq f(a) \geq f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be decreasing at $x = a$.

(1) When ' a ' be left end of the interval

$$f(a) \geq f(x) \forall x \in (a, a + h) \cap D_f \text{ for some } h > 0 \\ \Rightarrow f \text{ is decreasing at } x = a.$$

(2) When ' a ' be right end of the interval

$$f(x) \geq f(a) \forall x \in (a - h, a) \cap D_f \text{ for some } h > 0 \\ \Rightarrow f \text{ is decreasing at } x = a.$$

IV. Strictly decreasing at $x = a$:

If $f(s) > f(a) > f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be strictly decreasing at $x = a$.

(1) When ' a ' be left end of the interval

$$f(a) > f(x) \forall x \in (a, a + h) \cap D_f \text{ for some } h > 0 \\ \Rightarrow f \text{ is strictly decreasing at } x = a.$$

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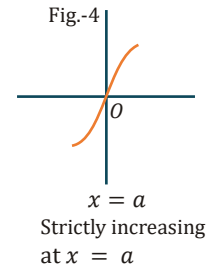
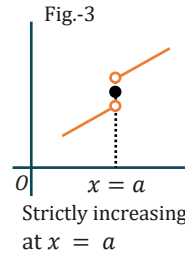
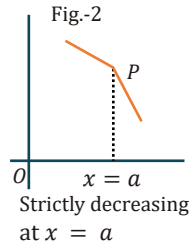
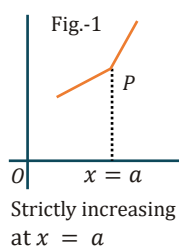


Illustration 1

Let $f(x) = x^3 - 3x + 2$. Examine the monotonicity of function at points $x = 0, 1$ & 2 .

Solution:

at $x = 0$,

$$f(0) = 2, f(0 + h) = h(h - 3) + 2 < 2$$

$$f(0 - h) = h(3 - h) + 2 > 2$$

$$\Rightarrow f(0 - h) > f(0) > f(0 + h)$$

f is strictly decreasing at $x = 0$.

at $x = 1$, $f(1 - h) > f(1) < f(1 + h)$ neither increasing nor decreasing

Similarly, at $x = 2$, $f(2 - h) < f(2) < f(2 + h)$ increasing at $x = 2$.

INCREASING DECREASING NATURE OF A FUNCTION IN AN INTERVAL:

Consider an interval $I \subseteq D_f$

I. Increasing Over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$, then f is increasing over interval I .

II. Decreasing Over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$, then f is decreasing over interval I .

III. Strictly increasing over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Leftrightarrow f(x_1) < f(x_2)$, then f is strictly increasing over interval I .

IV. Strictly decreasing over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Leftrightarrow f(x_1) > f(x_2)$, then f is strictly decreasing over interval I .

Monotonic Function:

If a function is either increasing or decreasing over an interval then it is said to be monotonic function over the interval.

If a function is either strictly increasing or strictly decreasing over an interval then it is said to be strictly monotonic function over the interval.

MONOTONICITY OF DIFFERENTIABLE FUNCTIONS:

For Differentiable Functions:

Consider an interval $I (\subseteq D_f)$ that can be $[a, b]$ or (a, b) or $[a, b)$ or $(a, b]$.

- (1) $f'(x) > 0 \forall x \in I \Rightarrow f$ is strictly increasing function over the interval I .
- (2) $f'(x) \geq 0 \forall x \in I \Rightarrow f$ is increasing function over the interval I .
- (3) $f'(x) > 0 \forall x \in I$ and $f'(x) = 0$ do not form any interval (that means $f'(x) = 0$ at discrete points)
 $\Rightarrow f$ is strictly increasing function over the interval I .
- (4) $f'(x) < 0 \forall x \in I \Rightarrow f$ is strictly decreasing function over the interval I .
- (5) $f'(x) \leq 0 \forall x \in I \Rightarrow f$ is decreasing function over the interval I .
- (6) $f'(x) \leq 0 \forall x \in I$ and $f'(x) = 0$ do not form any interval (that means $f'(x) = 0$ at discrete points)
 $\Rightarrow f$ is strictly decreasing function over the interval I .

Illustration 2

Prove that the function $f(x) = \log(x^3 + \sqrt{x^6 + 1})$ is strictly increasing.

Solution:

Now, $f(x) = \log(x^3 + \sqrt{x^6 + 1})$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6 + 1}} \left(3x^2 + \frac{6x^5}{2\sqrt{x^6 + 1}} \right) = \frac{3x^2}{\sqrt{x^6 + 1}} \geq 0$$

$\Rightarrow f(x)$ is strictly increasing.

EXAMPLES USING MONOTONICITY OF DIFFERENTIABLE FUNCTIONS:

Illustration 3

The function $f(x) = \tan x - x$

- (A) always increases
- (B) always decreases
- (C) never decreases
- (D) sometimes increases and sometimes decreases

Ans. (A)

Solution:

$$f(x) = \tan x - x$$

$$f'(x) = \sec^2 x - 1 > 0 \Rightarrow \text{Always increases.}$$

Illustration 4

Find the intervals of monotonicity of the function $y = x^2 - \log_e |x|$, ($x \neq 0$).

Solution:

$$\text{Let } y = f(x) = x^2 - \log_e |x|$$

$$f'(x) = 2x - \frac{1}{x} \quad ; \text{ for all } x (x \neq 0)$$

$$f'(x) = \frac{2x^2 - 1}{x} \Rightarrow f'(x) = \frac{(\sqrt{2}x - 1)(\sqrt{2}x + 1)}{x}$$

$$\text{So } f'(x) \geq 0 \text{ when } x \in \left[-\frac{1}{\sqrt{2}}, 0\right) \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \text{ and } f'(x) \leq 0$$

$$\text{when } x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup \left(0, \frac{1}{\sqrt{2}}\right]$$

$$\therefore f(x) \text{ is strictly increasing when } x \in \left[-\frac{1}{\sqrt{2}}, 0\right); \left[\frac{1}{\sqrt{2}}, \infty\right)$$

$$\text{and strictly decreasing when } x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right]; \left(0, \frac{1}{\sqrt{2}}\right]$$

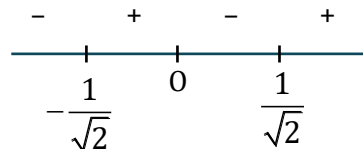


Illustration 5

If $g(x) = f(x) + f(1-x)$ and $f''(x) < 0$; $0 \leq x \leq 1$, show that $g(x)$ strictly increasing in $x \in (0, 1/2)$ and strictly decreasing in $x \in (1/2, 1)$

Solution:

$$\therefore f''(x) < 0 \Rightarrow f'(x) \text{ is strictly decreasing function.}$$

$$\text{Now, } g(x) = f(x) + f(1-x)$$

$$\therefore g'(x) = f'(x) - f'(1-x) \quad \dots(i)$$

Case I : If $x > (1-x) \Rightarrow x > 1/2$

$$\therefore f'(x) < f'(1-x) \Rightarrow f'(x) - f'(1-x) < 0$$

$$\Rightarrow g'(x) < 0$$

$$\therefore g(x) \text{ strictly decreases in } x \in \left(\frac{1}{2}, 1\right)$$

Case II : If $x < (1-x) \Rightarrow x < 1/2$

$$\therefore f'(x) > f'(1-x) \Rightarrow f'(x) - f'(1-x) > 0 \Rightarrow g'(x) > 0$$

$$\therefore g(x) \text{ strictly increases in } x \in (0, 1/2)$$

Illustration 6

Which of the following functions are strictly decreasing on $\left(0, \frac{\pi}{2}\right)$

(A) $\cos x$

(B) $\cos 2x$

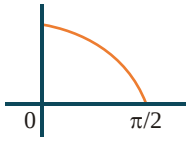
(C) $\cos 3x$

(D) $\tan x$

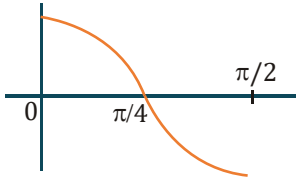
Ans. (A,B)

Monotonicity

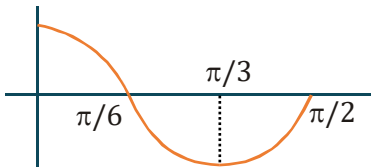
Solution:



$$f(x) = \cos x$$



$$f(x) = \cos 2x$$



$$f(x) = \cos 3x$$

Non-monotonic

$f(x) = \tan x$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$

Option A & B are correct.

Illustration 7

Prove that the equation $e^{(x-1)} + x = 2$ has one solution.

Solution:

$$\text{Let } f(x) = e^{(x-1)} + x$$

$$f'(x) = e^{(x-1)} + 1$$

$f(x)$ is strictly increasing function

$$\lim_{x \rightarrow \infty} f(x) = \infty \text{ \& \; } \lim_{x \rightarrow -\infty} f(x) = 0$$

$f(x) = 2$ has exactly one solution.

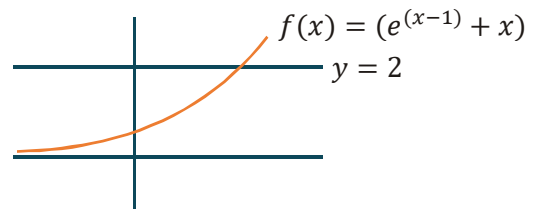


Illustration 8

The function f , defined by $f(x) = (x+2)e^{-x}$ is -

(A) decreasing for all x

(B) decreasing in $(-\infty, -1)$ and increasing $(-1, \infty)$ in

(C) increasing for all x

(D) decreasing in $(-1, \infty)$ and increasing in $(-\infty, -1)$

Solution:

$$f(x) = (x+2)e^{-x}$$

$$f'(x) = (x+2)e^{-x}(-1) + e^{-x} = -e^{-x}(x+1)$$

Increasing if $f'(x) > 0 \Rightarrow -e^{-x}(x+1) > 0$

$$\Rightarrow (x+1) < 0 \quad [\because e^{-x} \text{ always positive}]$$

$x \in (-\infty, -1)$ & decreasing in $(-1, \infty)$ as $f'(x) < 0$.

Illustration 9

Function $f(x) = x^3 + 6x^2 + (9+2k)x + 1$ is increasing function if

(A) $k \geq \frac{3}{2}$

(B) $k > \frac{3}{2}$

(C) $k < \frac{3}{2}$

(D) $k \leq \frac{3}{2}$

Ans. (A)

Solution:

$$\begin{aligned} f(x) &= x^3 + 6x^2 + (9 + 2k)x + 1 \\ f'(x) &= 3x^2 + 12x + (9 + 2k) \\ \Rightarrow 3x^2 + 12x + (9 + 2k) &\geq 0 \quad \forall x \in R \Rightarrow D \leq 0 \\ \Rightarrow 12 \cdot 12 - 12(9 + 2k) &\leq 0 \\ 3 - 2k &\leq 0 \\ \Rightarrow k &\geq \frac{3}{2}. \end{aligned}$$

Illustration 10

The function $f(x) = \cos x - 2px$ is monotonically decreasing for

- (A) $p < \frac{1}{2}$ (B) $p > \frac{1}{2}$ (C) $p < 2$ (D) $p > 2$

Ans. (A)

Solution:

$$\begin{aligned} f(x) &= \cos x - 2px \\ f'(x) &= -\sin x - 2p \\ \text{For monotonic decreasing } f'(x) &< 0 \\ \Rightarrow 2p &> -\sin x \\ \Rightarrow p &< \frac{1}{2} \quad [\because -1 \leq \sin x \leq 1] \end{aligned}$$

Illustration 11

The value of 'a' for which the function $f(x) = \sin x - \cos x - ax + b$ decreases for all real values of x , is

- (A) $a \geq -\sqrt{2}$ (B) $a \leq -\sqrt{2}$ (C) $a \leq \sqrt{2}$ (D) $a \geq \sqrt{2}$

Ans. (D)

Solution:

$$\begin{aligned} f(x) &= \sin x - \cos x - ax + b \\ f'(x) &= \cos x + \sin x - a \leq 0 \quad \forall x \in R \\ \Rightarrow a &\geq \cos x + \sin x \quad \forall x \in R \\ \Rightarrow a &\geq \sqrt{2} \end{aligned}$$

GREATEST AND LEAST VALUE OF FUNCTIONS:

- (a) If a continuous function $y = f(x)$ is increasing in the closed interval $[a, b]$, then $f(a)$ is the least value and $f(b)$ is the greatest value of $f(x)$ in $[a, b]$ (Figure-1)

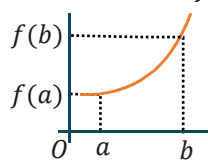


Figure-1

- (b) If a continuous function $y = f(x)$ is decreasing in $[a, b]$, then $f(b)$ is the least and $f(a)$ is the greatest value of $f(x)$ in $[a, b]$. (Figure-2)

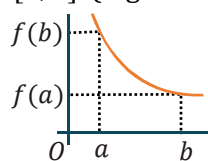


Figure-2

Monotonicity

- (c) If a continuous function $y = f(x)$ is increasing/decreasing in the (a, b) , then no greatest and least value exist.

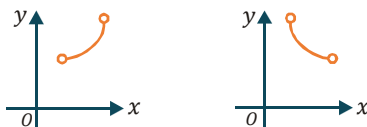


Illustration 12

Show that $f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x$ is strictly decreasing in $x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$. Also find its range.

Solution:

$$f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x = \tan^{-1} x - \ln x \Rightarrow f'(x) = \frac{1}{1+x^2} - \frac{1}{x} = \frac{-(1+x^2-x)}{x(1+x^2)}$$

$$\therefore f'(x) \leq 0 \quad \forall x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$$

$\Rightarrow f(x)$ is strictly decreasing.

$$f(x)|_{\max} = f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} + \frac{1}{2} \ln 3 \quad \& \quad f(x)|_{\min} = f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2} \ln 3$$

$$\therefore \text{Range of } f(x) = \left[\frac{\pi}{3} - \frac{1}{2} \ln 3, \frac{\pi}{6} + \frac{1}{2} \ln 3 \right]$$

Illustration 13

Find the minimum and maximum values of y in $4x^2 + 12xy + 10y^2 - 4y + 3 = 0$.

Solution:

$$(2x)^2 + 2 \cdot 2x \cdot 3y + (3y)^2 + (y-1)(y-3) = 0$$

$$\Rightarrow (2x+3y)^2 + (y-1)(y-3) = 0$$

$$x \in \mathbb{R}$$

$$\text{So } D \geq 0$$

$$144y^2 - 16(10y^2 - 4y + 3) \geq 0$$

$$16[-y^2 + 4y - 3] \geq 0$$

$$y^2 - 4y + 3 \leq 0$$

$$(y-1)(y-3) \leq 0$$

$$\therefore 1 \leq y \leq 3$$

$$\text{So } y_{\max} = 3 \text{ and } y_{\min} = 1$$

ESTABLISHING INEQUALITY:

Proving Inequalities using Monotonicity:

Comparison of two functions $f(x)$ and $g(x)$ can be done by analysing their monotonic behaviour.

Illustration 14

Let $f(x)$ and $g(x)$ are two function which are defined and differentiable for all $x \geq x_0$. If $f(x_0) = g(x_0)$ and $f'(x) > g'(x)$ for all $x > x_0$ then

(A) $f(x) < g(x)$ for some $x > x_0$

(B) $f(x) = g(x)$ for some $x > x_0$

(C) $f(x) > g(x)$ only for some $x > x_0$

(D) $f(x) > g(x)$ for all $x > x_0$

Ans. (D)

Solution:

$$\text{Let } h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x) > 0 \quad \forall x > x_0$$

$$h(x_0) = 0$$

$$\Rightarrow h(x) > 0 \quad \forall x > x_0$$

Illustration 15

For $x \in \left(0, \frac{\pi}{2}\right)$ prove that $\sin x < x < \tan x$

Solution:

$$\text{Let } f(x) = x - \sin x \Rightarrow f'(x) = 1 - \cos x$$

$$f'(x) > 0 \text{ for } x \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow f(x) \text{ is M.I. } \Rightarrow f(x) > f(0)$$

$$\Rightarrow x - \sin x > 0 \Rightarrow x > \sin x$$

Similarly consider another function $g(x) = x - \tan x \Rightarrow g'(x) = 1 - \sec^2 x$

$$g'(x) < 0 \text{ for } x \in \left(0, \frac{\pi}{2}\right) \Rightarrow g(x) \text{ is M.D.}$$

$$\text{Hence } g(x) < g(0)$$

$$x - \tan x < 0 \Rightarrow x < \tan x$$

$\sin x < x < \tan x$ Hence proved

Illustration 16

For $x \in (0, 1)$ prove that $x - \frac{x^3}{3} < \tan^{-1} x < x - \frac{x^3}{6}$.

Solution:

$$\text{Let } f(x) = x - \frac{x^3}{3} - \tan^{-1} x$$

$$\Rightarrow f'(x) = 1 - x^2 - \frac{1}{1+x^2} = -\frac{x^4}{1+x^2}$$

$$f'(x) < 0 \text{ for } x \in (0, 1)$$

$$\Rightarrow f(x) \text{ is M.D. } \Rightarrow f(x) < f(0)$$

$$\Rightarrow x - \frac{x^3}{3} - \tan^{-1} x < 0 \Rightarrow x - \frac{x^3}{3} < \tan^{-1} x \quad \dots(i)$$

$$\text{Similarly, } g(x) = x - \frac{x^3}{6} - \tan^{-1} x$$

$$g'(x) = 1 - \frac{x^2}{2} - \frac{1}{1+x^2} = \frac{x^2(1-x^2)}{2(1+x^2)}$$

$$g'(x) > 0 \text{ for } x \in (0, 1) \Rightarrow g(x) \text{ is M.I.}$$

$$x - \frac{x^3}{6} - \tan^{-1} x > 0 \Rightarrow x - \frac{x^3}{6} > \tan^{-1} x \quad \dots(ii)$$

from (i) and (ii), we get $x - \frac{x^3}{3} < \tan^{-1} x < x - \frac{x^3}{6}$ Hence Proved.

CONCAVITY – CONVEXITY OF FUNCTION:

Significance of the Sign of IInd Order Derivative:

The sign of the 2nd order derivative determines the concavity of the curve. If $f''(x) > 0 \forall x \in (a, b)$ then graph of $f(x)$ is concave upward in (a, b) . Similarly, if $f''(x) < 0 \forall x \in (a, b)$ then graph of $f(x)$ is concave downward in (a, b) .

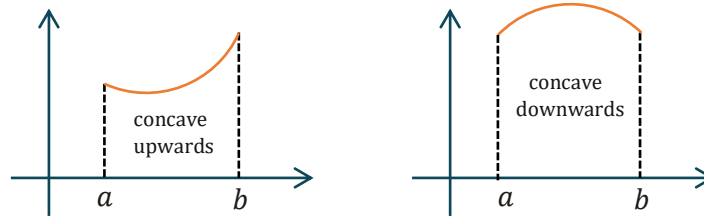


Illustration 17

Separate the intervals of concavity and convexity of $f(x) = x + \sin x$

Solution:

$$f'(x) = 1 + \cos x$$

$$f''(x) = -\sin x$$

$$f''(x) > 0 \forall x \in ((2n-1)\pi, 2n\pi) \quad n \in \mathbb{Z}$$

where graph is concave upward

$$f''(x) < 0 \forall x \in (2n\pi, (2n+1)\pi), n \in \mathbb{Z} \text{ Graph is concave downward.}$$

Illustration 18

If graph of $f(x) = 3x^4 + 2x^3 + ax^2 - x + 2$ is concave up for all real x , find values of a

Solution:

$$f''(x) = 36x^2 + 12x + 2a$$

$$f''(x) > 0 \Rightarrow 36x^2 + 12x + 2a > 0 \forall x \in \mathbb{R}$$

$$D < 0$$

$$\Rightarrow 12^2 - 4(36)(2a) < 0$$

$$1 - 2a < 0$$

$$a > \frac{1}{2}$$

ROLLE'S THEOREM:

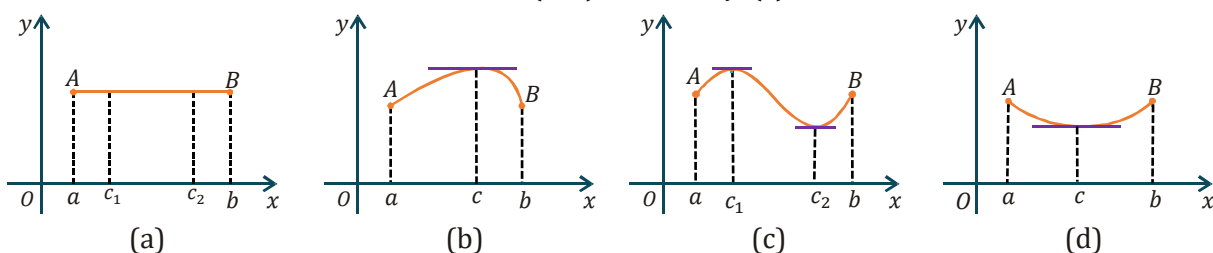
Let f be a function that satisfies the following three hypotheses:

(a) f is continuous on the closed interval $[a, b]$.

(b) f is differentiable on the open interval (a, b)

(c) $f(a) = f(b)$

Then there exist at least one number c in (a, b) such that $f'(c) = 0$.



Note :

If f is differentiable function then between any two consecutive roots of $f(x) = 0$, there is atleast one root of the equation $f'(x) = 0$.

(d) Geometrical Interpretation:

Geometrically, the Rolle's theorem says that somewhere between A and B the curve has at least one tangent parallel to x -axis.

Illustration 19

Verify Rolle's theorem for the function $f(x) = x^3 - 3x^2 + 2x$ in the interval $[0, 2]$.

Solution:

Here we observe that

(a) $f(x)$ is polynomial and since polynomial are always continuous, as well as differentiable. Hence $f(x)$ is continuous in the $[0, 2]$ and differentiable in the $(0, 2)$.

&

(b) $f(0) = 0, f(2) = 2^3 - 3 \cdot (2)^2 + 2(2) = 0$

$$\therefore f(0) = f(2)$$

Thus, all the condition of Rolle's theorem are satisfied.

So, there must exists some $c \in (0, 2)$ such that $f'(c) = 0$

$$\Rightarrow f'(c) = 3c^2 - 6c + 2 = 0 \Rightarrow c = 1 \pm \frac{1}{\sqrt{3}}$$

where both $c = 1 \pm \frac{1}{\sqrt{3}} \in (0, 2)$ thus Rolle's theorem is verified.

Illustration 20

Consider the function for $x \in [-2, 3]$, $f(x) = \begin{cases} \frac{x^3 - 2x^2 - 5x + 6}{x - 1} & \text{if } x \neq 1 \\ -6 & \text{if } x = 1 \end{cases}$, then

(A) f is discontinuous at $x = 1 \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$

(B) $f(-2) \neq f(3) \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$

(C) f is not derivable in $(-2, 3) \Rightarrow$ Rolle's theorem is not applicable

(D) Rolle's theorem is applicable as f satisfies all the conditions and c of Rolle's theorem is $\frac{1}{2}$

Ans. (D)

Solution:

$$f(-2) = f(3) = 0$$

$f(x)$ is continuous in $[-2, 3]$ & derivable in $(-2, 3)$ so Rolle's theorem is applicable.

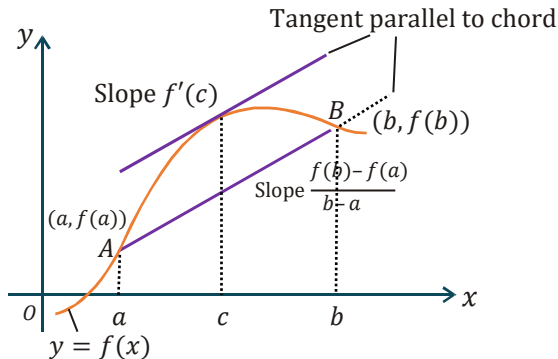
So, in $c \in (-2, 3)$ such that $f'(c) = 0$

$$\Rightarrow \frac{2c^3 - 5c^2 + 4c - 1}{(c-1)^2} = 0 \Rightarrow c = \frac{1}{2}$$

LAGRANGE'S MEAN VALUE THEOREM (LMVT):

Let f be a function that satisfies the following hypotheses:

- (i) f is continuous in $[a, b]$
- (ii) f is differentiable in (a, b) .



Then there is a number c in (a, b) such that

(a) Geometrical Interpretation:

Geometrically, the Mean Value Theorem says that somewhere between A and B the curve has at least one tangent parallel to chord AB .

(b) Physical Interpretations:

If we think of the number $(f(b) - f(a))/(b - a)$ as the average change in f over $[a, b]$ and $f'(c)$ as an instantaneous change, then the Mean Value Theorem says that at some interior point the instantaneous change must equal the average change over the entire interval.

Illustration 21

Find c of the Lagrange's mean value theorem for the function $f(x) = 3x^2 + 5x + 7$ in the interval $[1, 3]$.

Solution:

$$\text{Given } f(x) = 3x^2 + 5x + 7 \quad \dots(i)$$

$$\Rightarrow f(1) = 3 + 5 + 7 = 15 \text{ and } f(3) = 27 + 15 + 7 = 49$$

$$\text{Again } f'(x) = 6x + 5$$

$$\text{Here } a = 1, b = 3$$

Now from Lagrange's mean value theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow 6c + 5 = \frac{f(3) - f(1)}{3 - 1} = \frac{49 - 15}{2} = 17 \text{ or } c = 2.$$

Illustration 22

If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \leq f'(x) \leq 3$ for all x in $(-2, 5)$, then the greatest possible value of $f(5) - f(-2)$ is

- (A) 7 (B) 9 (C) 15 (D) 21

Ans. (D)

Solution:

Apply LMVT

$$f'(x) = \frac{f(5) - f(-2)}{5 - (-2)} \text{ for some } x \text{ in } (-2, 5)$$

$$\text{Now, } -4 \leq \frac{f(5) - f(-2)}{7} \leq 3$$

$$-28 \leq f(5) - f(-2) \leq 21$$

$\therefore$ Greatest possible value of $f(5) - f(-2)$ is 21.

GRAPH OF MISCELLANEOUS FUNCTION:

General Tips for Plotting the Graph:

- Compute points where the curve crosses the x -axis and also where it cuts the y -axis by putting $y = 0$ and $x = 0$ respectively.
- Compute $\frac{dy}{dx}$ and find the intervals where $f(x)$ is increasing or decreasing and also where it has horizontal tangent $\left(\frac{dy}{dx} = 0\right)$ or vertical tangent $\left(\frac{dy}{dx} \text{ not defined}\right)$.
- In regions where curve is monotonic compute y if $x \rightarrow \infty$ or $x \rightarrow -\infty$ to find whether y is asymptotic or not.
- If denominator vanishes say at $x = a$ then examine $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ to find whether f approaches ∞ or $-\infty$.

Illustration 23

$$f(x) = x^4 - 4x^2$$

Solution:

$$f(x) = x^4 - 4x^2 = 0$$

$$x^2(x^2 - 4) = 0$$

$$x = \pm 2, x = 0$$

$$f'(x) = 4x^3 - 8x = 0$$

$$= 4x(x^2 - 2)$$

$$\Rightarrow x = 0, x = \pm \sqrt{2}$$

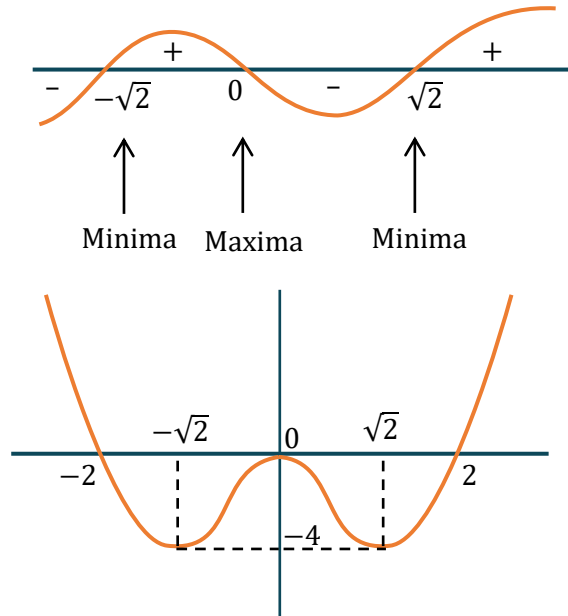


Illustration 24

$$f(x) = \frac{x}{\ln x} - 2$$

Solution:

$$f(x) = \frac{x}{\ln x} - 2 \quad \text{Domain of function} \Rightarrow x > 0, x \neq 1 \quad f'(x) = \frac{1}{\ln x} - \frac{x}{(\ln x)^2} \cdot \frac{1}{x} \Rightarrow \frac{\ln x - 1}{(\ln x)^2}$$

$$f'(x) = 0 \Rightarrow \ln x = 1$$

$$\Rightarrow x = e$$

